

数学物理方程(理科版)

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第五章 数学物理中的近似解法

§5.4 将微分方程化成差分方程

考虑

$$u'(x) = \lim_{\Delta x \rightarrow \infty} \frac{u(x+\Delta x) - u(x)}{\Delta x} = \lim_{\Delta x \rightarrow \infty} \frac{u(x) - u(x-\Delta x)}{\Delta x},$$

$$u''(x) = \lim_{\Delta x \rightarrow \infty} \frac{u'(x+\Delta x) - u'(x)}{\Delta x} = \lim_{\Delta x \rightarrow \infty} \frac{1}{\Delta x} \left[\frac{u(x+\Delta x) - u(x)}{\Delta x} - \frac{u(x) - u(x-\Delta x)}{\Delta x} \right]$$

$$= \lim_{\Delta x \rightarrow \infty} \frac{u(x+\Delta x) - 2u(x) + u(x-\Delta x)}{(\Delta x)^2}$$

当 $|\Delta x|$ 很小时, 分别取 $u'(x)$ 和 $u''(x)$ 的近似值如下:

$$u'(x) \approx \frac{u(x+\Delta x) - u(x)}{\Delta x} \quad \text{或} \quad u'(x) \approx \frac{u(x) - u(x-\Delta x)}{\Delta x}$$

$$u''(x) \approx \frac{u(x+\Delta x) - 2u(x) + u(x-\Delta x)}{(\Delta x)^2}$$

于是，微分方程就可以用差分方程近似替代。例如，

$$\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0,$$

用差分方程近似替代为

$$\frac{u(x + \Delta x, y) - 2u(x, y) + u(x - \Delta x, y)}{(\Delta x)^2} + \frac{u(x, y + \Delta y) - 2u(x, y) + u(x, y - \Delta y)}{(\Delta y)^2} = 0.$$

再利用Taylor展开式,

$$\begin{aligned}& u(x + \Delta x, y) - 2u(x, y) + u(x - \Delta x, y) \\&= \left[u(x, y) + \frac{\partial u(x, y)}{\partial x} \Delta x + \frac{1}{2!} \frac{\partial^2 u(x, y)}{\partial x^2} (\Delta x)^2 + \frac{1}{3!} \frac{\partial^3 u(x, y)}{\partial x^3} (\Delta x)^3 + \frac{1}{4!} \frac{\partial^4 u(\xi, y)}{\partial x^4} (\Delta x)^4 \right] \\&+ \left[u(x, y) - \frac{\partial u(x, y)}{\partial x} \Delta x + \frac{1}{2!} \frac{\partial^2 u(x, y)}{\partial x^2} (\Delta x)^2 - \frac{1}{3!} \frac{\partial^3 u(x, y)}{\partial x^3} (\Delta x)^3 + \frac{1}{4!} \frac{\partial^4 u(\eta, y)}{\partial x^4} (\Delta x)^4 \right] \\&- 2u(x, y) \\&= \frac{\partial^2 u(x, y)}{\partial x^2} (\Delta x)^2 + \frac{1}{4!} \left[\frac{\partial^4 u(\xi, y)}{\partial x^4} + \frac{\partial^4 u(\eta, y)}{\partial x^4} \right]\end{aligned}$$

其中 $\xi = x + \theta_1 \Delta x$, $\eta = x + \theta_2 \Delta x$ ($0 < \theta_1, \theta_2 < 1$), 于是有

$$\frac{u(x + \Delta x, y) - 2u(x, y) + u(x - \Delta x, y)}{(\Delta x)^2} = \frac{\partial^2 u(x, y)}{\partial x^2} + O((\Delta x)^2)$$

同理

$$\frac{u(x, y + \Delta y) - 2u(x, y) + u(x, y - \Delta y)}{(\Delta y)^2} = \frac{\partial^2 u(x, y)}{\partial y^2} + O((\Delta y)^2)$$

截断误差: $(\Delta x)^2 + (\Delta y)^2$ 。

类似地，一维热传导方程

$$\frac{\partial u}{\partial t} = a^2 \frac{\partial^2 u}{\partial x^2}$$

可以用差分方程近似替代为

$$\frac{u(x, t + \Delta t) - u(x, t)}{\Delta t} = a^2 \frac{u(x + \Delta, t) - 2u(x, t) + u(x - \Delta, t)}{(\Delta x)^2}$$

一维波动方程

$$\frac{\partial^2 u}{\partial t^2} = a^2 \frac{\partial^2 u}{\partial x^2}$$

可以用差分方程近似替代为

$$\begin{aligned} & \frac{u(x, t + \Delta t) - 2u(x, t) + u(x, t - \Delta t))}{(\Delta t)^2} \\ &= a^2 \frac{u(x + \Delta, t) - 2u(x, t) + u(x - \Delta x, t)}{(\Delta x)^2} \end{aligned}$$

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§5.5 偏微分方程的差分格式

1. Laplace方程的差分格式

1. Laplace方程的差分格式： Laplace方程的定解问题：

$$\begin{cases} \frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0 & (x, y) \in \Omega \\ u|_{\Gamma} = f(x, y) \end{cases}$$

做直线族：

$$\begin{aligned} x_i &= x_0 + ih \\ y_j &= y_0 + jh \end{aligned}$$

则

$$\left. \frac{\partial^2 u}{\partial x^2} \right|_{(x_i, y_j)} \approx \frac{u(x+\Delta x, y) - 2u(x, y) + u(x-\Delta x, y)}{(\Delta x)^2} \approx \frac{u(x_i+h, y_j) - 2u(x_i, y_j) + u(x_i-h, y_j)}{h^2}$$

$$\left. \frac{\partial^2 u}{\partial y^2} \right|_{(x_i, y_j)} \approx \frac{u(x, y+\Delta y) - 2u(x, y) + u(x, y-\Delta y)}{(\Delta y)^2} \approx \frac{u(x_i, y_j+h) - 2u(x_i, y_j) + u(x_i, y_j-h)}{h^2}$$

方程变为,

$$\begin{aligned}\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} &\approx \frac{u(x_i+h, y_j) - 2u(x_i, y_j) + u(x_i-h, y_j)}{h^2} + \frac{u(x_i, y_j+h) - 2u(x_i, y_j) + u(x_i, y_j-h)}{h^2} \\&= \frac{1}{h^2} [u(x_i+h, y_j) + u(x_i, y_j+h) + u(x_i-h, y_j) + u(x_i, y_j-h) - 4u(x_i, y_j)] \\&= 0\end{aligned}$$

即

$$\begin{aligned}u(x_i+h, y_j) + u(x_i, y_j+h) \\+ u(x_i-h, y_j) + u(x_i, y_j-h) - 4u(x_i, y_j) &= 0\end{aligned}$$

简记

$$U_{i+1,j} + U_{i,j+1} + U_{i-1,j} + U_{i,j-1} - 4U_{i,j} = 0$$

边界条件

$$u|_{(x_i, y_j) \in \Gamma_h} = f(x_i^*, y_i^*)$$

改写为：

$$U_{i,j} = \frac{U_{i+1,j} + U_{i,j+1} + U_{i-1,j} + U_{i,j-1}}{4}$$

$$(i = 0, 1, 2, \dots, m, j = 0, 1, 2, \dots, n)$$

同步迭代公式：

$$U_{i,j}^{(k+1)} = \frac{U_{i+1,j}^{(k)} + U_{i,j+1}^{(k)} + U_{i-1,j}^{(k)} + U_{i,j-1}^{(k)}}{4} \quad (k = 0, 1, 2, \dots)$$

异步迭代公式：

$$U_{i,j}^{(k+1)} = \frac{U_{i+1,j}^{(k)} + U_{i,j+1}^{(k)} + U_{i-1,j}^{(k+1)} + U_{i,j-1}^{(k+1)}}{4} \quad (k = 0, 1, 2, \dots)$$

迭代收敛条件:

$$\max_{i,j} \left| U_{i,j}^{(k+1)} - U_{i,j}^{(k)} \right| < \varepsilon$$

或者,

$$\frac{1}{N} \sum_{i,j} \left| U_{i,j}^{(k+1)} - U_{i,j}^{(k)} \right| < \varepsilon$$

2. 热传导方程的差分格式

考虑一维热传导方程的定解问题：

$$\begin{cases} \frac{\partial u}{\partial t} = a^2 \frac{\partial^2 u}{\partial x^2}, & 0 < x < 1, 0 < t < T \\ u|_{t=0} = f(x), & 0 < x < 1, f(0) = f(1) = 0 \\ u|_{x=0} = u|_{x=1} = 0 & 0 < t \leq T \end{cases}$$

做平行直线族：

$$x = x_i = i\Delta x \quad (i = 0, 1, 2, \dots, N),$$

$$t = t_j = j\Delta t \quad (j = 0, 1, 2, \dots, \left[\frac{T}{\Delta t}\right])$$

此处 $N\Delta x = 1$, $\left[\frac{T}{\Delta t}\right]$ 表示 $\frac{T}{\Delta t}$ 的整数部分。

在网格内的结点 (x_i, y_j) 处分别用

$$\frac{u(x_i, t_j) - u(x_i, t_j - \Delta t)}{\Delta t}$$

和

$$\frac{u(x_i + \Delta x, t_j - \Delta t) - 2u(x_i, t_j - \Delta t) + u(x_i - \Delta x, t_j - \Delta t)}{(\Delta x)^2}$$

代替导数

$$\frac{\partial u}{\partial t}, \quad \frac{\partial^2 u}{\partial x^2}$$

于是得到差分方程

$$\frac{U_{i,j} - U_{i,j-1}}{\Delta t} = a^2 \frac{U_{i+1,j-1} - 2U_{i,j-1} + U_{i-1,j-1}}{(\Delta x)^2}$$

和边界条件

$$\begin{cases} U_{i,0} = f(x_i), \end{cases}$$

令

$$\omega = a^2 \frac{\Delta t}{(\Delta x)^2}$$

则上面差分方程改写为

$$U_{i,j} = \omega(U_{i-1,j-1} + U_{i+1,j-1}) + (1 - 2\omega)U_{i,j-1}$$

可以证明，当

$$\omega = a^2 \frac{\Delta t}{(\Delta x)^2} \leq \frac{1}{2}$$

时，差分方程的解是稳定的，且收敛于原问题的解。

3. 波动方程的差分格式

考虑一维波动方程的定解问题：

$$\left\{ \begin{array}{l} \frac{\partial^2 u}{\partial t^2} = a^2 \frac{\partial^2 u}{\partial x^2} \quad 0 < x < 1, 0 < t < T \\ u|_{t=0} = \varphi(x) \quad 0 < x < 1, \quad \varphi(0) = \varphi(1) = 0 \\ \frac{\partial u}{\partial t}|_{t=0} = \psi(x) \\ u|_{x=0} = u|_{x=1} = 0 \quad 0 < t \leq T \end{array} \right.$$

做平行直线族：

$$x = x_i = i\Delta x$$

$$t = t_j = j\Delta t$$

在网格内的结点 (x_i, y_j) 处分别用

$$\frac{u(x_i, t_j + \Delta t) - 2u(x_i, t_j) + u(x_i, t_j - \Delta t))}{(\Delta t)^2}$$

和

$$\frac{u(x_i + \Delta x, t_j) - 2u(x_i, t_j) + u(x_i - \Delta x, t_j))}{(\Delta x)^2}$$

代替导数

$$\frac{\partial^2 u}{\partial t^2}, \quad \frac{\partial^2 u}{\partial x^2}$$

于是得到差分方程

$$\frac{U_{i,j+1} - 2U_{i,j} + U_{i,j-1}}{(\Delta t)^2} = a^2 \frac{U_{i+1,j} - 2U_{i,j} + U_{i-1,j}}{(\Delta x)^2}$$

和边界条件

$$\begin{cases} U_{i,0} = \varphi(x_i), \\ U_{i,1} - U_{i,0} = \psi(x_i)\Delta t, \\ U_{0,j} = 0, \quad U_{N,j} = 0 \end{cases}$$

令

$$\omega = a \frac{\Delta t}{\Delta x}$$

则上面差分方程改写为

$$U_{i,j+1} = \omega^2 (U_{i-1,j} + U_{i+1,j}) + 2(1 - \omega^2) U_{i,j} - U_{i,j-1}$$

可以证明，当

$$\omega = a \frac{\Delta t}{\Delta x} \leq 1$$

时，差分方程的解是稳定的，且收敛于原问题的解。

同学们，再见！